

CHANGXI ZHU

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605, Van Heuven Goedhartplein 13a, Utrecht

EDUCATION

Intelligent Systems, Utrecht University, The Netherlands

May 2021 - Present

Ph.D. in Artificial Intelligence

Thesis: Learning Tasks in Multi-agent Deep Reinforcement Learning with Communication.

Doctoral advisors: Prof. dr. Mehdi Dastani, Dr. Shihan Wang

South China University of Technology, China

2017-2020

Master in Software Engineering

Thesis: A Teacher-Student Framework for Knowledge Sharing in Multi-Agent Reinforcement Learning

Thesis supervisors: Prof. Yi Cai

Guangxi University, China

2013-2017

B.E. in Electronic Commerce

Thesis: Sentiment Analysis in Social Media Texts

Thesis supervisors: Prof. Jinchuang Zhao

WORKING EXPERIENCE

National University of Singapore

October 2020 - April 2021

Research Associate

- Develop an online multi-player collaborative decision-making environment using oTree, enabling real-time interaction between humans and RL agents; analyze behavioral data from both humans and RL agents.

Utrecht University

May 2021 - Present

Researcher

- Develop theories and algorithms for multi-agent deep reinforcement learning with communication, focusing on continuous, stochastic, and multi-task environments. Involved in teaching activities for the Master's program in Artificial Intelligence.

PUBLICATIONS

- **Changxi Zhu**, Mehdi Dastani, Shihan Wang. Learning Communication Skills in Multi-task Multi-agent Deep Reinforcement Learning. *arXiv preprint*, arXiv:2511.03348.
- **Changxi Zhu**, Mehdi Dastani, Shihan Wang. Reducing Variance Caused by Communication in Decentralized Multi-agent Deep Reinforcement Learning. *arXiv preprint*, arXiv:2502.06261.
- **Changxi Zhu**, Mehdi Dastani, Shihan Wang. Reducing Variance Caused by Communication in Decentralized Multi-agent Deep Reinforcement Learning. *Proceedings of the 24th International Conference on Autonomous Agents and MultiAgent Systems (AAMAS 2025)*, pp. 2841–2843.
- **Changxi Zhu**, Mehdi Dastani, Shihan Wang. Continuous Communication with Factorized Policy Gradients in Multi-agent Deep Reinforcement Learning. *Neural Computing and Applications*, pp. 1–24, 2025.
- **Changxi Zhu**, Mehdi Dastani, Shihan Wang. A Survey of Multi-agent Deep Reinforcement Learning with Communication. *Autonomous Agents and Multi-Agent Systems*, 38(1): 4, 2024.
- **Changxi Zhu**, Yi Cai, Shuyue Hu, Ho-fung Leung, Dickson K. W. Chiu. Learning by Reusing Previous Advice: A Memory-Based Teacher-Student Framework. *Autonomous Agents and Multi-Agent Systems*, 37(1): 14, 2023.

- **Changxi Zhu**, Ho-fung Leung, Shuyue Hu, Yi Cai. A Q-Values Sharing Framework for Multi-agent Reinforcement Learning under Budget Constraint. *ACM Transactions on Autonomous and Adaptive Systems*, 15(2): 4:1–4:28, 2021.
- **Changxi Zhu**, Yi Cai, Ho-fung Leung, Shuyue Hu. Learning by Reusing Previous Advice in Teacher-Student Paradigm. *Proceedings of the 19th International Conference on Autonomous Agents and MultiAgent Systems (AAMAS 2020)*, pp. 1674–1682.

TEACHING EXPERIENCE

Thesis Supervision at Utrecht University

- **BSc Artificial Intelligence**: Nikolaj Wilms (2023, *Emergence of a performant communication protocol in a multi-agent predator-prey game*); Frederik Makkinga (2024, *Automatic bias detection in fake news articles using machine learning*); Nemo ter Bogt (2025, *Implementation of a virtually trained UAV indoor navigation and collision avoidance model applicable in a physical 3D environment*)
- **MSc Artificial Intelligence**: Wouter Gerritsen (2024, *Learning Correlated Equilibrium without third party in MADRL using communication*)

Teaching Assistant at Utrecht University

- **MSc Artificial Intelligence**: Methods in AI Research (2021, 2022, 2023, 2024, 2025); Advanced Machine Learning (2021, 2022, 2023, 2024, 2025)

ACADEMIC SERVICE

Reviewers	ALA workshop@AAMAS (2022, 2023); TETCI (2023, 2025); NCAA (2024); NeurIPS (2024), ICLR (2025, 2026), ICML (2025), ECAI (2025), IBRL workshop@RLC (2025), AAAI (2026)
PC members	AAAI (2026)